THE SUPERFAST PROCESSORS

QUANTUM COMPUTERS HOLD OUT THE PROMISE OF EXPONENTIAL GROWTH IN PROCESSING POWER, ALLOWING COMPLEX COMPUTATIONS IN A FLASH. AND INDIA IS READY WITH A ROADMAP TO REALISE THE POTENTIAL OF THIS EMERGING TECHNOLOGY ACROSS SECTORS



GLOBAL LEADER
Sundar Pichai with one of Google's quantum computers in California, Oct. 2019

Developing a drug for a rare disease is a painstaking endeavour. You need to analyse millions of molecular combinations to find one that will work. Use a regular computer, and this could take years. But leverage quantum computing, and pharmaceutical firms can exponentially speed up complex molecular simulations to zero in on the most promising drug candidates in no time. This is the second quantum revolution. The first was in the early 1900s when new theories of quantum mechanics led to the later invention of pioneering technologies that we take for granted today—be it the laser, the MRI scanner, or even a photovoltaic cell. Now, quantum computing beckons as the next frontier.

WHY IT IS A GAME CHANGER

It's a very different realm, says Prof. Urbasi Sinha, who heads the Quantum Information and Computing (QuIC) laboratory at the Raman Research Institute in Bengaluru. "A quantum computer will not replace laptops, but achieve certain tasks exponentially faster," she adds. Hence, the global race to build them.

Quantum computing employs subatomic particles like trapped ions or photons to hold information. While a classical bit represents 0 or 1, a quantum bit (qubit) can be both simultaneously. This superposition multiplies possibilities manifold and theoretically enables a quantum computer to perform complex atomic-level simulations much faster than digital computers today—like modelling molecular combinations for drug discovery.

But even as further advancements in quantum computing hardware are awaited, other applications for quantum technologies are nearly here. Secure communications, for instance. Currently, quantum cryptography—which uses photons to deliver a quantum key—is being commercially tested to safeguard data connections. It could benefit sectors such as banking, defence and public health records, where securing data is

NATIONAL QUANTUM MISSION

Launched this year to scale up scientific and industrial R&D in quantum technology in India

₹6,004 CRORE
Total budget (FY24-FY31)

KEY TARGETS

- **50-100 qubit computers within 5 years; up to 1,000 qubit in 8 years**
- **Secure quantum communications: satellite-based, over 2,000 km, within India; long-distance with other countries**
- **Support design/synthesis of quantum materials such as superconductors**